

4. The oceans affect the biosphere by
 - (A) absorbing a substantial amount of the biosphere's oxygen.
 - (B) regulating the pH of freshwater biomes and terrestrial groundwater.
 - (C) decreasing winter temperatures of coastal terrestrial biomes.
 - (D) removing carbon dioxide from the atmosphere.
5. Which statement about dispersal is true?
 - (A) Dispersal is not a component of the life cycles of most plants and animals.
 - (B) Dispersal occurs only on an evolutionary time scale.
 - (C) Colonization of devastated areas after floods or volcanic eruptions depends on dispersal.
 - (D) The ability to disperse does not affect the geographic distribution of a species.
6. When climbing a mountain, we can observe transitions in biological communities that are analogous to the changes
 - (A) in biomes at different latitudes.
 - (B) in different depths in the ocean.
 - (C) in a community through different seasons.
 - (D) in an ecosystem as it evolves over time.
7. Suppose that the number of bird species is determined mainly by the number of vertical strata found in the environment. If so, in which of the following biomes would you find the greatest number of bird species?
 - (A) tropical rain forest
 - (B) savanna
 - (C) desert
 - (D) temperate broadleaf forest

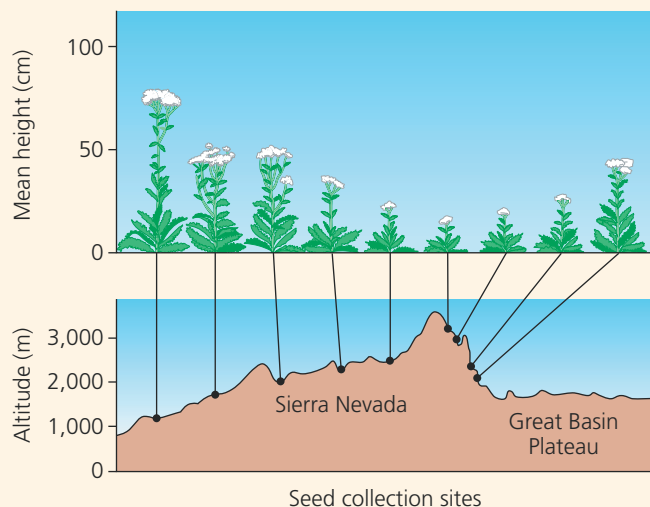
Levels 5-6: Evaluating/Creating

8. **WHAT IF?** If the direction of Earth's rotation reversed, the most predictable effect would be
 - (A) a big change in the length of the year.
 - (B) winds blowing from west to east along the equator.
 - (C) a loss of seasonal variation at high latitudes.
 - (D) the elimination of ocean currents.
9. **INTERPRET THE DATA** After examining Figure 52.20, you decide to study feeding relationships among sea otters, sea urchins, and kelp. You know that sea otters prey on sea urchins and that urchins eat kelp. At four coastal sites, you measure kelp abundance. Then you spend one day at each site and mark whether otters are present or absent every 5 minutes during the day. Graph kelp abundance (on the y-axis) versus otter density (on the x-axis), using the data below. Then formulate a hypothesis to explain any pattern you observe.

Site	Otter Density (# sightings per day)	Kelp Abundance (% cover)
1	98	75
2	18	15
3	85	60
4	36	25

10. **EVOLUTION CONNECTION** Discuss how the distribution of a species can be affected both by its evolutionary history and by ecological factors. Could ongoing evolutionary change also affect its distribution? Explain.

11. **SCIENTIFIC INQUIRY** Jens Clausen and colleagues, at the Carnegie Institution of Washington, studied how the size of yarrow plants (*Achillea lanulosa*) growing on the slopes of the Sierra Nevada varied with elevation. They found that plants from low elevations were generally taller than plants from high elevations, as shown in the diagram.



Data from J. Clausen et al., Experimental studies on the nature of species. III. Environmental responses of climatic races of *Achillea*, Carnegie Institution of Washington Publication No. 581 (1948).

Clausen and colleagues proposed two hypotheses to explain this variation within a species: (1) There are genetic differences between populations of plants found at different elevations. (2) The species has developmental flexibility and can assume tall or short growth forms, depending on local abiotic factors. If you had seeds from yarrow plants found at low and high elevations, how would you test these hypotheses?

12. **WRITE ABOUT A THEME: INTERACTIONS** Global warming is occurring rapidly in arctic marine and terrestrial ecosystems. The reflective white snow and ice cover are melting quickly and extensively, uncovering darker-colored ocean water, plants, and rocks. In a short essay (100–150 words), explain how this process might exemplify positive feedback.
13. **SYNTHESIZE YOUR KNOWLEDGE**



If you were to hike up Mount Kilimanjaro in Tanzania, you would pass through several habitats, including savanna at the base, forest on the slopes, and alpine tundra near the top. Explain how such diverse habitats can be found at one location near the equator.

For selected answers, see Appendix A.

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